



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL APPLICATIONS [SET A]

FULL MARKS: 100
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of six printed pages.*

SECTION A (40 Marks)
Attempt all questions from this Section

Question 1

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only)

- (i) Removal of top fertile layer of soil by external agents: [1]
(a) Soil pollution (b) Soil corrosion
(c) Soil erosion (d) Soil restoration
- (ii) Contour bunding is practiced mainly to [1]
(a) Increase irrigation efficiency
(b) Control soil erosion
(c) Improve soil fertility
(d) Reduce salinity
- (iii) The main beneficiaries of JFM are [1]
(a) Local communities (b) Private contractors
(c) State forest department (d) Industrial sectors
- (iv) Expired bottle of Dettol is an example of [1]
(a) Domestic waste (b) Hazardous waste
(c) Biomedical waste (d) E waste
- (v) Which of the following is NOT a traditional method of water harvesting? [1]
(a) Check dams (b) Bore wells
(c) Eris (d) Tankas

(vi)



[1]

Which environmental principle does the picture best reflect?

- (a) Extended producer responsibility
- (b) Community participation
- (c) Polluter pay
- (d) Green GDP

(vii)

Assertion: Wind energy plants cannot operate efficiently at low wind speeds.

[1]

Reason: They depend on kinetic energy of moving air for turbine rotation.

- (a) Both assertion and reason are true and reason is the correct explanation of assertion
- (b) Both assertion and reason are true and reason is not the correct explanation of assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(viii)

Assertion: Natural Resource Accounting is a process of estimating the value of natural resource depletion and environmental degradation due to economic activities.

[1]

Reason: NRA just estimates the value of natural resources.

- (a) Both assertion and reason are true and reason is the correct explanation of assertion
- (b) Both assertion and reason are true and reason is not the correct explanation of assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(ix)

The residents of Rampur village decided to set up a community biogas plant to manage cattle dung and kitchen waste. The plant now supplies clean cooking fuel to 25 households and produces nutrient-rich slurry used as manure in nearby fields. This initiative has significantly reduced the use of firewood and minimized smoke-related health issues among women.

[1]

The nutrient-rich slurry obtained from the biogas plant is an example of

- (a) Air pollutant
- (b) Renewable energy
- (c) By product of waste recycling
- (d) Industrial residue

- (x) **Cleaner Production Technology focuses on** [1]
 (a) Managing waste after production
 (b) Preventing waste during production
 (c) Transporting waste safely
 (d) Dumping waste in landfills
- (xi) **A scientific facility decides to launch a remote sensing satellite. Where will it be placed?** [1]
 (a) Laboratories (b) Underground
 (c) Outer space (d) In a submarine
- (xii) **Effective segregation of waste at the point of generation is very important. Select the appropriate statements giving the importance of waste segregation.** [1]
 (i) Less waste goes to the landfill
 (ii) Help in reducing waste
 (iii) Better for public health and environment
 (iv) Waste pickers spend less time sorting through garbage
 (a) i and ii (b) i and iii (c) ii and iv (d) i and iv
- (xiii) **Which of the following statements best describe the working principle of an Electrostatic Precipitator (ESP) in controlling air pollution?** [1]
 (a) It uses high-voltage electric fields to charge and collect dust particles from exhaust gases before releasing clean air.
 (b) It burns particulate matter at high temperatures to reduce its volume.
 (c) It filters pollutants through activated charcoal to absorb harmful gases.
 (d) It sprays water on exhaust gases to wash out suspended impurities.
- (xiv) **A major advantage of Euro VI/Bharat Stage VI norms is** [1]
 (a) Reduced vehicular emission
 (b) Cheaper fuel
 (c) Higher engine noise
 (d) Use of leaded petrol
- (xv) **Select the correct sequence of EIA process** [1]
 (i) Scoping
 (ii) Impact analysis
 (iii) Screening
 (iv) Mitigation
 (a) i, iii, ii, iv (b) iii, i, ii, iv (c) ii, iv, iii, i (d) iv, ii, i, iii
- (xvi) **The residue of sugarcane used for making paper is called _____.** [1]
 (a) Husk (b) Bagasse
 (c) Pulp (d) Coir

- (xvii) Dioxin is an example of toxic gas produced during [1]
 (a) Mining (b) Composting
 (c) Incineration (d) Construction
- (xviii) Select the incorrect pair: [1]
 (a) Leachate – Composting
 (b) Freon – CFC
 (c) NGO – NABARD
 (d) Landfill-- Methane
- (xix) Fluidized bed boilers work on [1]
 (a) Solid waste (b) Liquid waste
 (c) Gaseous waste (d) Both solid and liquid
- (xx) What is the primary purpose of a check dam? [1]
 (a) To store water for irrigation and prevent flooding
 (b) To create reservoirs for large cities
 (c) To generate electricity through hydroelectric power
 (d) To purify water in rural areas

Question 2

- (i) List two environmental hazards of e-waste disposal. [2]
 (ii) *The next World War would be fought for water.* Justify the statement. [2]
 (iii) What is Dastkar? Enumerate its objectives. [3]
 (iv) A village plans to switch from fuel wood to biogas plants for cooking. Identify the health and environmental problems that will be mitigated due to this transformation. [3]

Question 3

- (i) What do you mean by End of Pipe treatment? [2]
 (ii) Identify the problems of solid waste management in India. [2]
 (iii) [3]

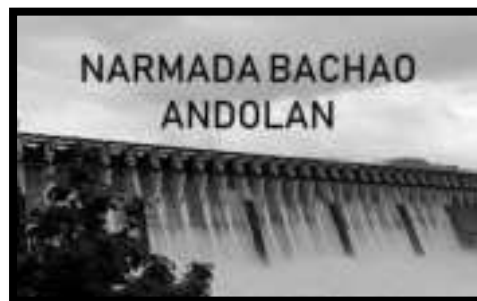


- (iv) Identify the technology. Mention the characteristics of the product obtained. [3]
 Mention any three aims of NGOs working for women welfare.

SECTION B (60 MARKS)
(Attempt any four questions from this section)

Question 4

- (i) Study the picture and answer the questions that follows: [5]



- (ii) Why do environmentalists usually protest the construction of big dams? [5]
(iii) Give an account of bubble theory. [5]
Smart phones, computers and television are few examples of modern technology. [5]
Mention any three advantages and two disadvantages of modern technology.

Question 5

- (i) The East Kolkata Wetlands in West Bengal are one of India's most important natural ecosystems. These wetlands treat nearly 900 million litres of Kolkata's wastewater everyday through natural processes. However, rapid urbanization and illegal dumping of waste have reduced wetland area. [5]
(a) Mention the importance of wetlands.
(b) What threats will be there if wetlands decline?
(ii) Explain how the work initiated by Tarun Bharat Sangh changed the life of people in Rajasthan. [5]
(iii) How does globalisation affect the environment and society? [5]

Question 6

- (i) Name the founders of Gandhigram. Mention three objectives of Gandhigram. [5]
(ii) What is dry compost toilet? Give four advantages of using dry compost toilets. [5]
(iii) Write a short note on Curitiba's traffic management. [5]

Question 7

- (i) [5]



- Identify the phenomenon and discuss the causes associated with it.
- (ii) Mention the steps that can be taken to prevent deforestation. [5]
- (iii) Consumption of contaminated water with sewage and effluent is the root cause of many water borne diseases. [5]
- Name the process that should be practiced to prevent the contamination of water.
- Mention four advantages of the above process.

Question 8

- (i) A community in Tamil Nadu wanted to conserve water for the benefit of the society. They came up with a particular plan of action in sync with the place's traditions. What according to you is their plan? Give a rough sketch of the structure. Mention two benefits expected out of this plan. [5]
- (ii) As a member of the school eco club, how will you ensure that resources are conserved in school by students, teachers and office staff? [5]
- (iii) In August 2023, heavy rainfall caused severe floods in the state of Assam. Many villages were submerged, and thousands of people had to be rescued. The government used satellite images and drones to monitor the condition. What is remote sensing? How did it help to monitor the condition? [5]

Question 9

- (i) Water is the elixir of life. Your science teacher wants you to prepare a project plan for assessment. 'How to save water?' [5]
- (a) Write any three ways that you will suggest to bring awareness in your neighbourhood.
- (b) Recommend any two methods to prevent further depletion of the underground water table.
- (ii) Why should we recycle water? [5]
- (iii) "Environmental experts have raised concerns over the yellowing of the Taj Mahal's white marble. Rising air pollution from vehicles, industrial emissions, and the burning of waste in Agra have contributed to acid rain and soot deposition on the monument's surface. Authorities have also strengthened pollution control measures in the Taj Trapezium Zone (TTZ) to protect the heritage site." [5]
- (Source: The Times of India, July 12, 2023)
- Mention three steps taken by the Mathura oil refinery and two steps taken by the Delhi government to address this issue.